The logo for the UC Berkeley Center for Human-Compatible AI. It features the text "UC Berkeley" in a bold, dark font, with "Center for Human-Compatible AI" in a lighter, sans-serif font below it, all enclosed within a light gray rectangular border.

UC Berkeley Center for Human-Compatible AI

This is the inaugural issue of the [Center for Human-Compatible AI](#)'s quarterly newsletter. CHAI's mission is to develop the conceptual and technical foundations required to build provably beneficial AI systems. CHAI was inaugurated in fall 2016 and is funded by the Open Philanthropy Project, the Future of Life Institute, the Center for Long-Term Cybersecurity, and the Leverhulme Trust. CHAI investigators include faculty members at Berkeley, Cornell, and Michigan.

In this issue we'll highlight key research from our faculty and students, round up recent media spotlights, and share insights from our 2017 Annual Workshop.

Research highlights

Obedience, off-switches, and cooperative inverse reinforcement learning

Last year, PhD student Dylan Hadfield-Menell and his advisors formalized the [cooperative inverse reinforcement learning](#) (CIRL) problem, in which a human and robot are both rewarded according to the human's reward function, but the robot starts off not knowing what this reward function is. Since CIRL makes it in both players' best interests for the robot to learn how to act in a way that fulfills the human's reward function, this setup leads to desirable cooperative learning behaviors such as active teaching, active learning, and communicative actions.

This year, meaningful progress has been made within the CIRL setup.

[Should Robots be Obedient?](#) (Milli et al., 2017) explores the tradeoffs of robots being perfectly obedient vs. exhibiting independent judgment. The research finds that when a human is imperfectly rational and a robot attempts to infer the human's underlying preferences rather than being perfectly obedient, the robot performs its function better than it would by following the human's orders precisely. In these situations, having a well-specified model of the features the human cares about or the human's level of rationality are essential for the robot to exhibit good judgment in strategically disobeying orders.

In the example below (extreme for the purpose of illustration): the human wants a slice of cake, but in a slip of the tongue, asks for a vial of poison instead. In this case, obedience becomes harmful.

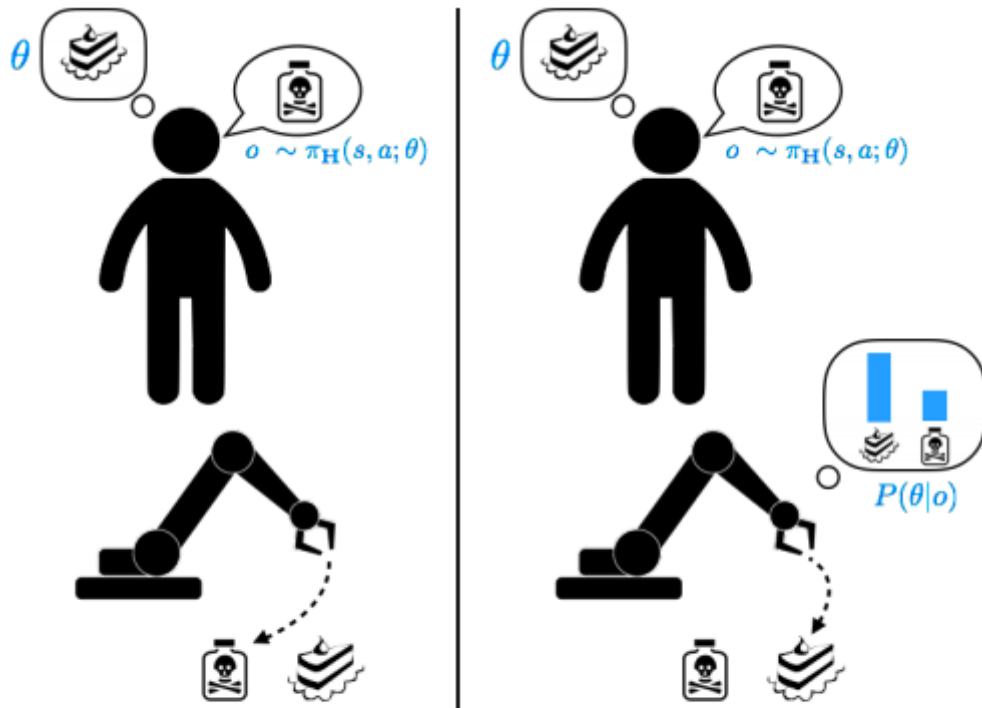


Figure 1: (Left) The blindly obedient robot always follows H's order. (Right) An IRL-R computes an estimate of H's preferences and picks the action optimal for this estimate.

[Milli et al., 2017](#)

[The Off-Switch Game](#) (Hadfield-Menell et al., 2017) presents meaningful progress on the corrigibility problem. An intuitive desideratum for a corrigible AI system is the ability for a human to turn the system off without the system interfering. The off-switch game extends the CIRL setup: we have a robot R with an off switch that it can disable, and a human H that can press the robot's off switch. The idea of a human-usable off-switch is difficult to preserve, since reward-maximizing agents will typically have strong incentives for self-preservation. But in a CIRL environment where both H and R are maximizing a shared reward function and only H knows what that reward function looks like, R can only update its prior beliefs about the reward function by observing H's actions. If H wants to shut down R, and R doesn't know the reward parameters, then R will want this to happen, as a way to prevent errors arising from its own lack of knowledge!

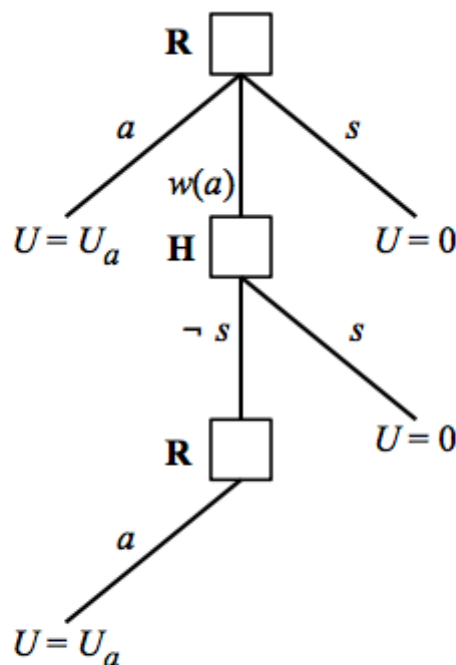


Figure 1: The structure of the off-switch game. Squares indicate decision nodes for the robot **R or the human **H**.**

a = take an action, s = hit the off switch, $w(a)$ = wait for the human to decide. ([Hadfield-Menell et al., 2017](#))

The key insight is that the more uncertain **R** is about the reward function (i.e. the more variance there is in **R**'s prior), the more corrigible it is -- it is more amenable to letting **H** switch it off. The intuition is simple: if **R** believes that **H** is a rational actor, and **R** is uncertain about the reward function, then **R** is more likely to defer to **H** in the decision to hit the off switch. On the other hand, if **R** believes it has precise knowledge of the reward function, then it knows exactly what it should do. This might include proceeding with action a and preventing the human from hitting the off switch. If we can have this failsafe in our back pocket, we should be very happy. See the talk [here](#).

Of course, a perfect “CIRL solver” robot can still fail to be corrigible—that is to say, resist being shut down in a detrimental way—if its engineers have misspecified the CIRL problem that it's supposed to solve, for example, with an incorrect model of the human's rationality, or an incorrect model of how the robot should update its beliefs about human preferences over time. Thus, while we believe CIRL presents a valuable lense through which to view the problem of corrigibility, a great deal of valuable follow-up work remains to be done in order to more comprehensively address the potential for engineering errors to yield incorrigible systems.

Causality, intention, and responsibility

One objective of CHAI's current research is to formalize established concepts in social sciences that lack the logical and mathematical precision required for incorporating these concepts into the design of intelligent agents.

Causality and blame are evaluated daily in courts of law and have long been debated in the context of thought experiments like the Trolley Problem -- but as we develop autonomous agents in the real world that must necessarily make moral judgments in their operation, we will require a more specific way of defining the extent to which an agent's actions (causes) are responsible, or blameworthy, for certain outcomes (effects).

Joe Halpern's recent research at Cornell focuses on this topic. When formalizing a heuristic for autonomous vehicles like "avoid harming humans," how do we incorporate probabilities into tradeoffs (e.g. .01% chance of harming a person vs. 99.9% chance of creating massive property damage)? How do we determine whether an agent's actions are indeed responsible for certain benefit or harm done, and the extent to which they are responsible when many actors are involved? A prerequisite to implementing any moral decision-making framework is having a way to define causality and quantify responsibility. Halpern's book [Actual Causality](#) (2016) and his [recent talk](#) go into more detail on these topics.

2017 Annual Workshop recap

The Center for Human-Compatible Artificial Intelligence (CHAI) hosted its first annual workshop on Friday, May 5 and Saturday, May 6, 2017. Topics covered include preference aggregation, single-principal value alignment, interactive control, and minimum-viable world-takeover capabilities. See more details [here](#).

CHAI assorted links: news, articles, videos

- Stuart Russell, [Three principles for creating safer AI](#), TED talk, Vancouver, April 25, 2017.
- [The future is now](#), debate organized by Wadham College, Oxford, in San Francisco, June 5, 2017.
- Stuart Russell, [AI for Good](#), interview at the UN Global Summit on AI for Good, June 9, 2017.
- Stuart Russell, [Provably Beneficial Artificial Intelligence](#) (starts at 55.25), keynote at IJCAI 2017 ([slides here](#)).
- [As Tech Titans Bicker, Should You Worry About the Rise of AI?](#), by Ryan Nakashima and Matt O'Brien, Top Tech News, July 30, 2017.
- Stuart Russell, [Do we want machines to make decisions on killing in war?](#), interview on BBC Tomorrow's World, June 1, 2017.
- [Top US general warns against rogue killer robots, by John Bowden, The Hill, July 18, 2017.](#)

For inquiries, feedback, or requests for future content, please contact chai-newsletter-editor@humancompatible.ai.

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